# Md. Jaied Hasan

Md. Jaied Hasan | jh-emon002 | mdjaiedhasan02@gmail.com | +8801722656638

## Research Profile

Electrical and Electronic Engineering graduate from Bangladesh University of Engineering and Technology (BUET), majoring in Communication and Signal Processing, with current research on Wi-Fi 8 multi-AP coordination, coordinated beamforming, and interference analysis. Experienced in wireless network simulation, signal processing, machine learning, and privacy-preserving federated analytics.

## Education

**Bangladesh University of Engineering and Technology (BUET), Dhaka** — Jan 2022 – Jun 2026
B.Sc. in Electrical and Electronic Engineering — **CGPA: 3.80/4.00** (Top 27%)
**Major:** Communication and Signal Processing — *Last four semesters: 3.85/4.00*

## Research Interests

- Multi-AP Coordination and Spatial Reuse
- PHY/MAC Interference and Propagation Modeling
- Wireless Measurements and Network Intelligence
- Data-Driven Wireless Resource Allocation
- Wireless Simulation and Experimental Validation

## Research Experience

**Research Assistant – Department of EEE, BUET** — Sep 15, 2026 – Present
*Supervisor: Dr. Md. Forkan Uddin, Professor, EEE, BUET*

- Investigating Wi-Fi 8 (IEEE 802.11bn) multi-AP coordination and coordinated beamforming (Co-BF), with emphasis on inter-cluster interference, throughput, and spatial reuse.
- Proposed a DRL based solution to the interference caused by multiple AP cluster deployments.
- Developed a Python-based wireless simulator comparing CSMA/CA with beamforming, Co-BF with inter-cluster interference, and an idealized interference-free Co-BF reference case.
- Implemented channel modeling, propagation loss and phase, ZF precoding, carrier sensing, contention/backoff, and omnidirectional NAV signaling.
- Analyzed per-AP throughput, SINR, beam patterns, and spatial SINR heatmap; extending evaluations toward larger 10-AP/30-STA deployments and event-level MAC analysis.

**Undergraduate Thesis: Federated Sparse PCA with Differential Privacy Using Correlated Noise**
*Supervisor: Dr. Hafiz Imtiaz, Professor, EEE, BUET*

- Developed a federated sparse-PCA framework for horizontally partitioned datasets and investigated correlated-noise mechanisms for differentially private dimensionality reduction.
- Evaluated privacy–utility, sparsity, convergence, reconstruction trade-offs using reproducible Python experiments.
- Manuscript based on the thesis is currently under review.

## Other Experience

**FlyRank AI** — Aug 1, 2026 – Present
*Machine Learning Intern* — *Remote*

- Developed an ML pipeline to detect and rank declining web content across 23,000+ webpages using historical traffic and trend-based features.
- Performed feature engineering, Precision@K evaluation, model diagnostics, time/group-aware validation, and data-leakage audits.

## Selected Relevant Projects

**ML-Based Mobile Traffic Forecasting and Dynamic Bandwidth Planning**

- Developed a regional mobile-internet traffic forecasting pipeline using BTCL data; benchmarked XGBoost, GRU, and LSTM models using MAE/RMSE.
- Investigated how traffic predictions could inform network-demand forecasting and dynamic bandwidth/capacity planning.

**HIVE-R / GPS-Free Rescue Drone for Disaster Response**

- Developed a semi-autonomous rescue drone for GPS-denied environments with IMU-assisted navigation and dual-camera visual odometry.
- Integrated Raspberry Pi-based human detection and wireless reporting to a ground-control station, combining sensing, edge inference, and communication.

**LifeSaver: IoT-Based Smart Helmet and Emergency Response System**

- Built an ESP32-based prototype integrating an IMU, GPS, GSM communications, and a mobile application for crash detection and emergency alerts.

## Research Outputs

- **Journal manuscript under review:** Federated sparse PCA with differential privacy using a correlated-noise mechanism.
- **Conference paper accepted (2026):** *An Integrated Dual-Module Cyber Physical System for Motorcycle Safety: Utilizing Pre-Ride Safety, Crash Detection and Swift Response with Mobile Application Companion.*
- **Conference submission (2026):** ML-based mobile traffic forecasting for network-capacity planning.

## Technical Skills

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|---|---|
| **Programming** | Python, C, C++, MATLAB, Verilog |
| **Wireless / Signal Processing** | Wi-Fi 8, multi-AP coordination, Co-BF, DSP |
| **Machine Learning** | PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas |
| **Research Methods** | Wireless simulation, statistical evaluation, time-series forecasting, feature engineering, federated learning, differential privacy, sparse PCA |
| **Embedded / Tools** | ESP32, Raspberry Pi, Arduino, ESP32-CAM, GPS/GSM, IMU sensors, Linux, Git/GitHub, MATLAB/Simulink, ROS |

## Selected Coursework

Wireless Communications • Communication Systems I & II • Linear Algebra • Radar and Satellite Communications • Digital Signal Processing • Probability and Statistics • Artificial Intelligence and Machine Learning

## Honors and Awards

- **Dean's List Scholarship** – Level 1 and Level 3; **University Merit Scholarship**.
- **Shamsul Huda Chowdhury Scholarship**; **Professor Dr. Ahmad Ullah and Mrs. Chemanara Ahmad Ullah Trust Fund Scholarship**.
- **1st Runner-Up, Brain of Varendra** – talent-hunt competition organized by Varendra University, Rajshahi.